Finite-Element Drift-Diffusion Modelling of Direct-Gap III–V Light-Emitting Diodes: Wavelength Tuning and Double-Heterostructure Efficiency in Nano-Photonic Emitters

Author One*, Author Two†
Department of Electronics and Communication Engineering
Affiliation, City, Country
*email@domain †email@domain

***Abstract*—Direct-gap III–V semiconductors are the material basis of nano-photonic light emitters, and their design—band-gap engineering for a target wavelength and heterostructure engineering for efficiency—requires a device simulator that couples carrier transport to radiative recombination and optical output. We present an open-source, two-dimensional finite-element drift-diffusion (DD) solver that adds a bimolecular (radiative) recombination channel and an optical post-processor reporting the internal quantum efficiency (IQE), emission wavelength and optical power, together with heterojunction band offsets (Anderson's rule), a self-consistent Schrödinger–Poisson quantum correction, and Fermi–Dirac statistics for degenerate cladding. For a GaAs homojunction LED the solver reproduces the analytic built-in potential (1.151 V, numeric $=$ analytic) and an emission wavelength of 873 nm set by the 1.42 eV gap, with a peak IQE of 0.53. Sweeping the active material tunes the emission from the near-infrared (GaAs, 873 nm) through the red ($Al_{0.3}Ga_{0.7}As$, 689 nm) to the ultraviolet (GaN, 366 nm), following the material band gap. A GaAs/AlGaAs double-heterostructure (DH) LED confines both carrier types in the narrow-gap active region; the resulting overlap raises the peak IQE from 0.53 (homojunction) to 0.90, quantifying the classic DH efficiency advantage on a single, self-consistent transport model. The tool provides a compact, physically transparent platform for the design of nano-scale direct-gap emitters.**

***Index Terms*—Light-emitting diode, radiative recombination, direct-gap semiconductor, double heterostructure, internal quantum efficiency, drift-diffusion, finite-element method, GaAs, Al-GaAs, GaN, nano-photonics.**

I. Introduction

LIGHT-EMITTING diodes built from direct-gap III–V semiconductors are central to solid-state lighting, displays, optical interconnect and sensing, and their nanoscale active regions—thin double heterostructures, quantum wells—are the emerging materials of advanced photonic devices [1]. Two engineering levers dominate their design: the choice of active material, which sets the emission wavelength through the band gap, and the heterostructure, which confines injected carriers to raise the radiative efficiency [2], [3]. Predictive design therefore needs a device model that couples electrical transport to the radiative process and to the optical output.

The drift-diffusion (DD) framework [4] is the standard basis for such modelling. To describe an emitter it must be augmented with (i) a radiative (bimolecular) recombination channel that both consumes carriers and produces photons, (ii) heterojunction band-edge offsets so that a narrow-gap active region clad by a wider-gap material actually confines carriers, and (iii)—for the thin, degenerately-doped layers of a real device—quantum-confinement and Fermi–Dirac corrections. This paper presents a compact, open-source finite-element solver that integrates all of these, and uses it to characterise wavelength tuning across III–V materials and the efficiency gain of a double-heterostructure emitter.

Contributions. (i) A 2-D finite-element DD solver with a bimolecular radiative channel and an optical post-processor (IQE, wavelength, optical power) that is current-consistent by construction; (ii) heterojunction electrostatics, a self-consistent Schrödinger–Poisson correction and Fermi–Dirac statistics on the same mesh; (iii) validated GaAs LED characteristics; (iv) demonstrations of material-tuned emission from the near-infrared to the ultraviolet and of the double-heterostructure efficiency advantage.

II. Device Model

A. Drift-diffusion core

The steady state is solved in the normalised quasi-Fermi-potential variables $u = \psi/V_T$, $v = \phi_n/V_T$, $w = \phi_p/V_T$ (De Mari scaling [5]), with $n = n_i e^{u-v}$, $p = n_i e^{w-u}$. Poisson's equation and the two carrier continuity equations are discretised with linear (P1) triangular finite elements and assembled into a $3N \times 3N$ system solved by a fully-coupled damped Newton method using an exact analytic Jacobian. Terminal currents are extracted as the sum of the natural continuity residuals over a contact's nodes, which is conservative by construction.

B. Radiative recombination and optical output

The light-emitting mechanism of a direct-gap diode is radiative (bimolecular) recombination, added as a net-recombination term

$$R_{\text{rad}} = B\,(np - n_i^2), \quad (1)$$

where B ($\text{m}^3\,\text{s}^{-1}$) is the material radiative coefficient. Because of the $(np - n_i^2)$ prefactor the rate vanishes at equilibrium and leaves it unchanged; its exact derivatives $\partial R/\partial n = Bp$, $\partial R/\partial p = Bn$ enter the analytic Jacobian, preserving quadratic convergence. Shockley–Read–Hall and Auger channels are included as competing non-radiative paths. After a converged forward-bias solve the optical post-processor integrates the radiative rate over the device (on the same quadrature and corrected densities used in the assembly) and reports

$$I_{\text{rad}} = q\,R_{\text{rad}}, \qquad \text{IQE} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{SRH}} + R_{\text{Auger}}}, \quad (2)$$

$$E_{\text{ph}} = E_g^{\text{active}}, \qquad \lambda = \frac{hc}{E_{\text{ph}}}, \quad P_{\text{opt}} = R_{\text{rad}} E_{\text{ph}}. \quad (3)$$

By construction qR_{rad} equals the radiative part of the terminal current, so the electrical and optical outputs are mutually consistent. The radiative coefficient is significant only in direct-gap materials; the bundled direct-gap files (GaAs, AlGaAs, GaN) carry representative 300 K values, and a guardrail warns if radiative emission is requested on an indirect material.

C. Heterojunction, quantum and Fermi–Dirac corrections

Multi-material devices are solved with a heterojunction electrostatics core: position-dependent permittivity and mobility, and band-edge offsets $\Delta E_c, \Delta E_v$ from Anderson's electron-affinity rule that appear as fixed additive shifts in the carrier-density exponents. A narrow-gap active region clad by a wider-gap material therefore presents potential barriers that confine both carrier types. For thin bodies a self-consistent Schrödinger–Poisson correction quantises the transverse states into subbands and enters transport as a charge-conserving additive quantum potential; for degenerately-doped cladding, Fermi–Dirac statistics enter as a degeneracy shift refreshed by the same outer self-consistency loop. All corrections vanish in their respective non-degenerate, bulk limits, recovering the classical Boltzmann solver exactly.

III. Results and Discussion

A. GaAs LED: validation and characteristics

A GaAs p–n homojunction ($N_A = N_D = 10^{16}\,\text{cm}^{-3}$, $E_g = 1.42$ eV, $n_i = 2.15 \times 10^6\,\text{cm}^{-3}$, $B = 2 \times 10^{-16}\,\text{m}^3\text{s}^{-1}$) gives a numerical built-in potential of 1.151 V, matching the analytic $V_{bi} = V_T \ln(N_A N_D / n_i^2) = 1.151$ V. Fig. 1 shows the forward I–V together with the radiative component $I_{\text{rad}} = qR_{\text{rad}}$; at 1.3 V the device carries 53.7 μA total with 1.68 μA radiative and 2.4 μW of optical output. The emission

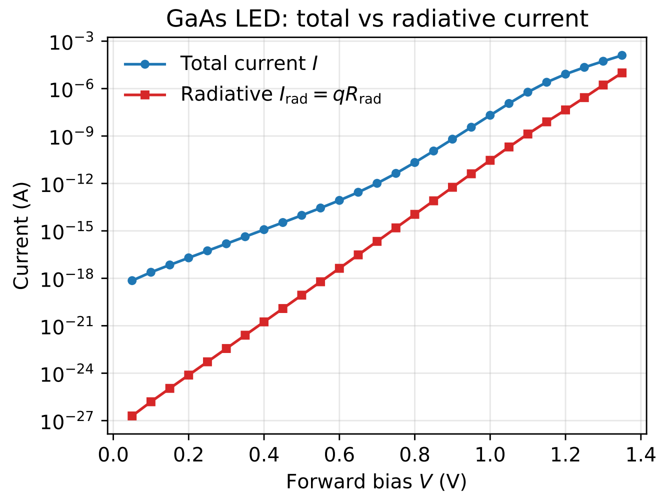

Fig. 1. GaAs LED forward characteristics: total terminal current and its radiative part $I_{\text{rad}} = qR_{\text{rad}}$, which are consistent by construction.

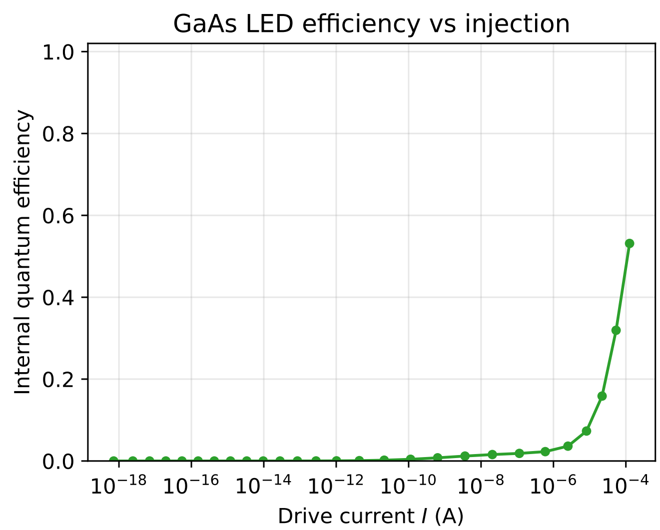

Fig. 2. GaAs LED internal quantum efficiency versus drive current. IQE rises as the bimolecular radiative rate overtakes non-radiative SRH recombination, peaking at 0.53.

wavelength is 873 nm ($E_{\text{ph}} = 1.42$ eV), the expected near-infrared line of GaAs. The internal quantum efficiency (Fig. 2) rises with injection as the bimolecular rate ($\propto np$) overtakes the SRH channel, reaching a peak of 0.53.

B. Material-tuned emission

Keeping the device geometry fixed and changing only the active material tunes the emission wavelength through the band gap (Fig. 3, Table I): GaAs emits at 873 nm (near-infrared), $Al_{0.3}Ga_{0.7}As$ at 689 nm (red), and GaN at 366 nm (ultraviolet). The solver converges the full electrical problem for each material—spanning intrinsic densities from $n_i \sim 10^6\,\text{cm}^{-3}$ (GaAs) down to the extremely small n_i of wide-gap GaN—and returns the emission line directly from the active-region gap.

TABLE I
MATERIAL-TUNED EMISSION (FIXED GEOMETRY, $N_A = N_D = 10^{16}$ CM^{-3})

active material	E_g (eV)	λ (nm)	spectral band
GaAs	1.42	873	near-infrared
$Al_{0.3}Ga_{0.7}As$	1.80	689	red (visible)
GaN	3.39	366	ultraviolet

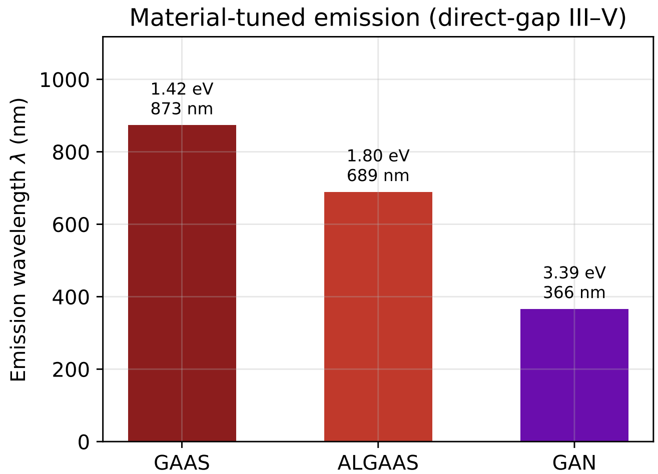

Fig. 3. Emission wavelength for three direct-gap III–V active materials, labelled with the band gap. The wavelength follows $\lambda = hc/E_g$, spanning the near-infrared to the ultraviolet.

This spans the technologically important range from optical-communication wavelengths to visible and UV emitters on a single, consistent model.

C. Double-heterostructure efficiency advantage

Fig. 4 shows the carrier densities along the transport direction of a GaAs/AlGaAs double-heterostructure (DH) LED: a 100 nm GaAs active region ($\sim 10^{16}\,\mathrm{cm}^{-3}$) clad on both sides by degenerately-doped ($\sim 10^{18}\,\mathrm{cm}^{-3}$) AlGaAs, with a 10 nm body solved with the heterojunction, Schrödinger–Poisson and Fermi–Dirac options all enabled. The Anderson band offsets confine *both* electrons and holes to the narrow-gap active region, where their densities rise by orders of magnitude and overlap—the physical basis of efficient recombination in a real DH emitter. The device built-in potential is 1.766 V and the emission remains at 873 nm, correctly originating from the GaAs active region despite the AlGaAs cladding.

The efficiency consequence is quantified in Fig. 5: confining the carriers raises the peak IQE to 0.90, compared with 0.53 for the GaAs homojunction of Fig. 2—the classic double-heterostructure advantage, reproduced end-to-end on a single self-consistent transport model. The same framework thus captures both design levers of a nano-photonic emitter: wavelength through the active material and efficiency through the heterostructure.

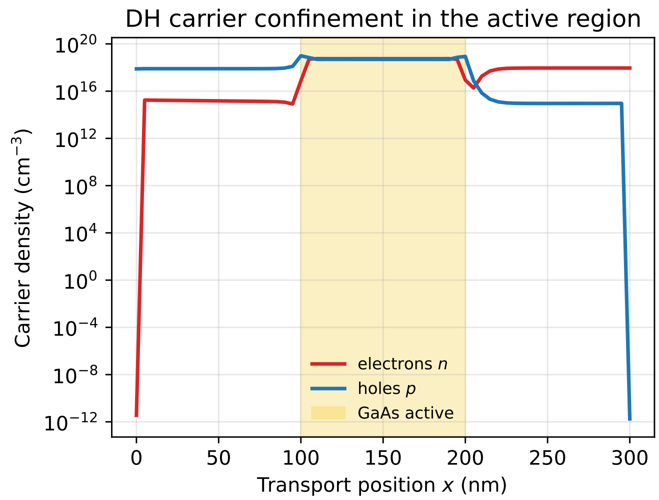

Fig. 4. Carrier confinement in the GaAs/AlGaAs double-heterostructure LED (shaded: GaAs active region). Heterojunction band offsets confine both electrons and holes to the active region, where they overlap and recombine radiatively.

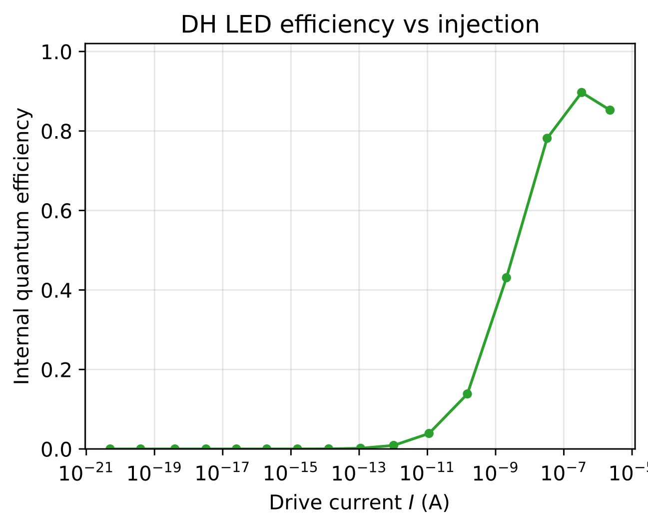

Fig. 5. Internal quantum efficiency of the double-heterostructure LED versus drive current. Carrier confinement raises the peak IQE to 0.90, versus 0.53 for the GaAs homojunction (Fig. 2).

D. Discussion

Three features make the model suitable for nano-photonic device design. First, the radiative rate shares the terminal-current scale, so qR_{rad} is exactly the radiative part of the current and the optical output cannot drift from the electrical solution. Second, all corrections are opt-in and vanish in their bulk limits, so a single code path spans a simple homojunction and a fully-corrected DH quantum device. Third, the analytic Jacobian is preserved by every correction, keeping the solve robust. The present optical model reports internal quantum efficiency and the active-gap emission line; extraction efficiency, spectral broadening and multi-quantum-well active regions (kp subbands) are natural extensions.

IV. Conclusion

We have presented an open-source finite-element drift-diffusion solver for direct-gap III–V light-emitting diodes that couples carrier transport to bimolecular radiative recombination and a current-consistent optical post-processor, with heterojunction, Schrödinger–Poisson and Fermi–Dirac corrections on the same mesh. The GaAs LED characteristics are validated against analytic physics; material selection tunes the emission from 873 nm (GaAs) to 366 nm (GaN); and a GaAs/AlGaAs double heterostructure raises the peak internal quantum efficiency from 0.53 to 0.90 by confining carriers to the active region. The tool is a compact, physically transparent platform for the design of nanoscale direct-gap emitters.

Software and Reproducibility

The solver and the example scripts that generated every figure are open-source (MIT licence) and require only a structured mesh built by the bundled generator.

References

[1] E. F. Schubert, *Light-Emitting Diodes*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2006.

[2] H. Kroemer, "A proposed class of heterojunction injection lasers," *Proc. IEEE*, vol. 51, no. 12, pp. 1782–1783, 1963.

[3] H. C. Casey and M. B. Panish, *Heterostructure Lasers*. New York: Academic Press, 1978.

[4] S. Selberherr, *Analysis and Simulation of Semiconductor Devices*. Vienna: Springer-Verlag, 1984.

[5] A. De Mari, "An accurate numerical steady-state one-dimensional solution of the p-n junction," *Solid-State Electron.*, vol. 11, no. 1, pp. 33–58, 1968.

[6] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed. Hoboken, NJ: Wiley, 2007.

[7] R. L. Anderson, "Experiments on Ge–GaAs heterojunctions," *Solid-State Electron.*, vol. 5, no. 5, pp. 341–351, 1962.